



SAFETY Department
SIMHADRI

Note No :1



Ref:SMTTPP/SAFETY/2023-24/LMI /363677

Date:04/08/2023(09:29:40)

Subject:Approval of LMI on EMERGENCY RESPONSE PLANNING -reg.

Sub:- Approval of LMI on Emergency Response Planning -reg.

The LMI on Emergency Responce Planning, Ref no: LMI -OGN - OPS-SAFT - 002 , Dated: Aug -2023 is reviewed and no changes incorporated in the LMI.

This Supersedes LMI -OGN - OPS-SAFT - 002 , Dated: Jul -2021.

The LMI is attached for reference.

Put up for approval of Competent authority.

Submitted by:

पुली सदाशिव रेड्डी - उप महाप्रबंधक

Puli Sadashiva Reddy - 062135 (DY. GENERAL MANAGER ,SAFETY) ,Simhadri
(PSADASHIVAREDDY@NTPC.CO.IN, 9425823806)

On:04/08/2023(09:29:40) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:2

Submitted for approval.

Submitted by:

अर्जुन प्रदीप नायर - अपर महाप्रबंधक

Arjun Pradeep Nair - 004419 (ADDL. GENERAL MANAGER ,TECH SERVICES)
(APNAIR@NTPC.CO.IN, 9403964693) ,Simhadri

On:04/08/2023(10:52:22) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:3

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

Uma Shanker Verma - 006137 (ADDL. GENERAL MANAGER ,EEMG)
(USVERMA@NTPC.CO.IN, 9650992821) ,Simhadri

On:04/08/2023(11:47:51) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:4

Annexure-1 may be reviewed as compared to CC document

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:04/08/2023(12:01:24) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:5

Reviewed

Submitted by:

पुली सदाशिव रेड्डी - उप महाप्रबंधक

Puli Sadashiva Reddy - 062135 (DY. GENERAL MANAGER ,SAFETY)
(PSADASHIVAREDDY@NTPC.CO.IN, 9425823806) ,Simhadri

On:07/08/2023(11:10:10) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:6

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:07/08/2023(11:12:08) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:7

Review pl.

Submitted by:

राजेश चावला - अपर महाप्रबंधक

Rajesh Chawla - 007383 (ADDL. GENERAL MANAGER ,MTP)
(RAJESHCHAWLA@NTPC.CO.IN, 9425534528) ,Simhadri

On:07/08/2023(11:19:36) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:8

Reviewed. No changes are proposed. Forwarded for approval.

Submitted by:

भास्कर वी के कुरीचेटी - उप महाप्रबंधक

Bhaskar V K Kurichety - 008045 (DY. GENERAL MANAGER ,MTP)
(KBHASKAR@NTPC.CO.IN, 9445864820) ,Simhadri

On:07/08/2023(12:10:07) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:9

Submitted by:

राजेश चावला - अपर महाप्रबंधक

Rajesh Chawla - 007383 (ADDL. GENERAL MANAGER ,MTP)
(RAJESHCHAWLA@NTPC.CO.IN, 9425534528) ,Simhadri

On:07/08/2023(15:37:54) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:10

Forwarded

Submitted by:

इंद्रजीत जेना - वरिष्ठ विशेषज्ञ

Indrajeet jena - 097748 (SR. SPECIALIST ,PAEDIATRICIAN)
(INDRAJEETJENA@NTPC.CO.IN, 9437001049) ,Simhadri

On:07/08/2023(18:07:45) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:11

please review

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:08/08/2023(15:57:11) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:12

Reviewed , forwarded for approval.

Submitted by:

तन्मय सिन्हा - वरिष्ठ प्रबंधक

TANMOY SINHA - 101612 (SENIOR MANAGER ,ASH HANDLING)
(TANMOYSINHA@NTPC.CO.IN, 9434737629) ,Simhadri

On:09/08/2023(09:48:07) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:13

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:09/08/2023(11:02:58) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:14

Forwarded for approval.

Submitted by:

डॉ वी जयन - अपर महाप्रबंधक

Dr Velayudhanpillai Jayan - 006202 (ADDL. GENERAL MANAGER ,ENVT. MGT.)
(VJAYAN@NTPC.CO.IN, 9650991721) ,Simhadri

On:09/08/2023(11:50:00) **Ref:**SMTPP/SAFETY/2023-24/LMI /363677

Note No:15

Reviewed and submitted for approval.

Submitted by:

जी एच राव - अपर महाप्रबंधक

G H Rao - 006589 (ADDL. GENERAL MANAGER ,C & I MAINT- BOILER)
(GHNRAO@NTPC.CO.IN, 8331016245) ,Simhadri

On:09/08/2023(15:37:48) **Ref:**SMTPP/SAFETY/2023-24/LMI /363677

Note No:16

Submitted by:

Amulya Kumar Behera - 097433 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)
(AMULYABEHERA@NTPC.CO.IN, 9425534368) ,Simhadri

On:12/08/2023(10:50:15) **Ref:**SMTPP/SAFETY/2023-24/LMI /363677

Note No:17

Submitted by:

ASIS KUMAR MUKHOPADHYAY - 004586 (ADDL. GENERAL MANAGER ,MECH MAINT- OFF SITE)

(AKMUKHOPADHYAY@NTPC.CO.IN, 9434038935) ,Simhadri

On:12/08/2023(11:18:29) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:18

Pl review.

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)

(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri

On:12/08/2023(11:30:49) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:19

LMI reviewed. Please forward.

Submitted by:

लवण्या कलत्थुरु - प्रबंधक

Lavanya Kalatthuru - 103064 (MANAGER ,CHEMISTRY)

(LAVANYAK@NTPC.CO.IN, 7382621063) ,Simhadri

On:14/08/2023(16:56:33) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:20

Reviewed and forwarded for approval.

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)
(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri

On:14/08/2023(17:06:52) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:21

Reviewed and Forwarded for approval

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER)
(OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri

On:14/08/2023(23:34:12) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:22

Review

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

Shambhu Kumar Suman - 005007 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(SKSUMAN@NTPC.CO.IN, 9431215296) ,Simhadri

On:16/08/2023(16:10:45) **Ref:**SMTPP/SAFETY/2023-24/LMI /363677

Note No:23

Reviewed. Forwarded for further approval.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

RASHMI RANJAN SATAPATHY - 100837 (SENIOR MANAGER ,MECH MAINT- TURBINE)
(RRSATPATHY@NTPC.CO.IN, 9437021584) ,Simhadri

On:21/08/2023(17:08:12) **Ref:**SMTPP/SAFETY/2023-24/LMI /363677

Note No:24

Reviewed & forwarded for approval pl.

Submitted by:

आन्नप्रगद रमा कोटेश्वरराव - अपर महाप्रबंधक

Annapragada Rama Koteswararao - 047211 (ADDL. GENERAL MANAGER ,FUEL HAND.- OPN)
(ARKOTESWARARAO@NTPC.CO.IN, 8332988907) ,Simhadri

On:23/08/2023(15:34:44) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:25

Reviewed and forwarded for approval pl.

Submitted by:

सन्जित मजूमदार - अपर महाप्रबंधक

Sanjit Majumdar - 005065 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)
(SMAZUMDAR@NTPC.CO.IN, 9650990065) ,Simhadri

On:23/08/2023(15:47:17) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:26

Submitted by:

प्रसन्न कुमार के एस - अपर महाप्रबंधक

K S Prasanna Kumar - 004504 (ADDL. GENERAL MANAGER ,IT)
(PRASANNAKMR@NTPC.CO.IN, 9428822389) ,Simhadri

On:23/08/2023(17:02:49) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:27

Reviewed and forwarded for approval.

Submitted by:

कासा वेंकटेश्वर प्रसाद - अपर महाप्रबंधक

Kasa Venkateswara Prasad - 006486 (ADDL. GENERAL MANAGER ,REGIONAL LEARNING INSTITUTE)

(KVPRASAD02@NTPC.CO.IN, 9650997388) ,Simhadri

On:24/08/2023(18:14:23) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:28

Reviewed and forwarded for approval.

Submitted by:

रूमा डे शर्मा - अपर महाप्रबंधक

Ruma De Sharma - 006923 (ADDL. GENERAL MANAGER ,HR)

(RDSHARMA@NTPC.CO.IN, 9650999329) ,Simhadri

On:25/08/2023(11:17:04) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:29

Please review

Submitted by:

देबजीत कुमार कर - अपर महाप्रबंधक

Debjit Kumar Kar - 003722 (ADDL. GENERAL MANAGER ,C&M)
(DEBJITKAR@NTPC.CO.IN, 9650993524) ,Simhadri

On:25/08/2023(12:38:20) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:30

Please review

Submitted by:

अशफुल हक - वरिष्ठ प्रबंधक

ASRAFUL HOQUE - 041919 (SENIOR MANAGER ,STORES)
(ASRAFULHAQUE@NTPC.CO.IN, 9491396765) ,Simhadri

On:25/08/2023(15:47:32) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:31

Reviewed .May please forward

Submitted by:

मधु बाबू आर्वा - प्रबंधक

Madhu Babu Arava - 035200 (MANAGER ,STORES)
(MBARAVA@NTPC.CO.IN, 9493906039) ,Simhadri

On:28/08/2023(10:12:49) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:32

Noted and forwarded

Submitted by:

देबजीत कुमार कर - अपर महाप्रबंधक

Debjit Kumar Kar - 003722 (ADDL. GENERAL MANAGER ,C&M)
(DEBJITKAR@NTPC.CO.IN, 9650993524) ,Simhadri

On:28/08/2023(10:25:09) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:33

Submitted by:

जगनमोहन राव कोनेटी - महाप्रबंधक

Koneti Jaganmohan Rao - 005969 (GENERAL MANAGER ,FGD)
(KJRAO@NTPC.CO.IN, 9650998443) ,Simhadri

On:28/08/2023(18:43:38) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:34

Pl go through the proposed LMI especially w.r.to the responsibilities mentioned for Operation Dept/SCEs and if found in line with the OGN may further forward for approval

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

TR Rajagopalan - 003994 (GENERAL MANAGER ,OPERATION)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:01/09/2023(09:18:01) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:35

Reviewed and is forwarded for apprpval.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,OPERATION)
(MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri

On:01/09/2023(16:00:46) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:36

Please attach the DMP in soft for perusal. Also make it a part of the LMI.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:11/09/2023(09:07:20) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:37

As discussed, DMP (Onsite emergency plan) is attached sir.

Submitted by:

पुली सदाशिव रेड्डी - उप महाप्रबंधक

Puli Sadashiva Reddy - 062135 (DY. GENERAL MANAGER ,SAFETY)
(PSADASHIVAREDDY@NTPC.CO.IN, 9425823806) ,Simhadri

On:13/09/2023(15:48:29) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:38

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:13/09/2023(18:22:45) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:39

Forwarded for approval please.

Submitted by:

धुरजति प्रसाद पात्रा - महाप्रबंधक

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,O & M)
(DPPATRA@NTPC.CO.IN, 8275045446) ,Simhadri

On:14/09/2023(00:52:31) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

Note No:40

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

Sanjay Kumar Sinha - 005185 (GENERAL MANAGER ,HEAD OF PROJECT)
(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

On:14/09/2023(09:58:59) **Ref:**SMTTPP/SAFETY/2023-24/LMI /363677

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

Local Management Instruction – LMI-OGN/OPS/SAFT/002

Revision: 00, Dt: AUG 2023

EMERGENCY RESPONSE PLANNING
OF
SIMHADRI SUPER THERMAL POWER STATION

Approved for Implementation by.....
HEAD OF PROJECT (Simhadri)

Date: __.08.2023

SAFETY DEPARTMENT

TITLE OF LMI:		Emergency Response Planning of Simhadri Super Thermal Power Station					
LMI No:		<i>LMI-OGN-OPS-SAFT-002</i>					
ISSUE NO	REVISION NO	DATE	CHANGES DONE	PREPARED BY	CHECKED BY	REVIEWED BY	APPROVED BY
<i>01</i>	<i>00</i>	<i>.07.21</i>	<i>All clauses are new as the document is being issued for the first time.</i>	<i>Safety Officer</i>	<i>ALL HODs</i>	<i>GM(O&M)</i>	<i>HOP</i>
<i>02</i>	<i>00</i>	<i>.08.23</i>	<i>Nil</i>	<i>Safety Officer</i>	<i>ALL HODs</i>	<i>GM(O&M)</i>	<i>HOP</i>

N T P C L I M I T E D

SIMHADRI SUPER THERMAL POWER STATION

LMI APPROVAL DOCUMENT

Title of LMI:	Emergency Response Planning of Simhadri Super Thermal Power Station.	
LMI No.:	LMI-OGN-OPS-SAFT-002	
Issue No. : 2	Revision: 0	Date : .08.2023

	Name Sri.	Designation	Signature
Prepared by	P Sada Shiva Reddy	DGM (Safety)	
Checked by	Arjun Pradeep Nair	AGM (TS & Safety)	
	Uma Shankar Verma	AGM (EEMG)	
	Rajesh Chawla	AGM(MTP)	
	Dr. Indrajeet Jena	Senior Specialist	
	Nilanjan Mondal	AGM(AHP)	
	Dr. V Jayan	AGM (EMG)	
	G Hari Narayana Rao	AGM (C&I)	
	Amulya Kumar Behera	AGM(FT & FC)	
	Asis Kumar Mukhopadhyay	AGM(OFS)	
	Piyush Srivastava	AGM (Chemistry	
	Om Prakash	AGM (BMD)	
	Shambhu Kumar Suman	AGM (TMD)	
	A R Koteswar Rao	AGM(CHP -OPN)	
	Sanjit Majumdar	AGM(CHP-Maint))	
	K S Prasanna Kumar	AGM(IT)	
	K V Prasad	AGM(RLI)	
	Ruma De Sharma	AGM (HR)	
	Debjit Kumar Kar	AGM (C&M Stores)	
	K Jaganmohan Rao	GM (FGD)	
	T R Rajagopalan	GM(OPN)	
	Mathew Eipe Kovoov	Gm(Maint)	
Reviewed by	D P Patra	GENERAL MANAGER (O&M)	
Approved for Issue by	Sanjay Kumar Sinha	GROUP GENERAL MANAGER (S)	
Date of Issue			

INDEX

SL NO	CONTENTS	PAGE NO
1.0	Scope	5
2.0	Superseded documents	5
3.0	References	5
4.0	Revision Details	5
5.0	Introduction	5
6.0	Roles and Accountability	6
7.0	Requirements	7
7.1	Identification of Emergency Scenarios	7
7.2	Emergency Management Plan	7
7.3	Emergency Communication	8
7.4	Emergency Response Teams	9
7.5	Emergency Coordinator	9
7.6	Control Center/ Emergency Equipment	9
7.7	Maintenance of Emergency Equipment	10
7.8	Headcount of Persons within the Premises	10
7.9	Exercises	11
8.0	Records	11
9.0	Audit	12
10.0	Review	13
Annexure - 1	Emergency Scenarios.	14
Annexure - 2	Outline of Emergency.	15

1.0 SCOPE :

Emergency at a location is an adverse event which if not managed effectively, could seriously impact the location with a highly undesirable outcome that can cripple its business potential for an unspecified period of time. Disaster is a type of Emergency.

This LMI provides guidance to develop and maintain Emergency Response Plans (ERP), also referred to as Disaster Management Plans (DMP) or Emergency Management Plans (EMP) or Emergency Action Plans (EAP), for the protection of personnel and property of NTPC and the general public and for the recovery from emergencies and disasters in the shortest possible time in all locations where the operational responsibility and ownership lies with the company.

This LMI covers broad types of locations and scenarios and primarily focuses on safety aspects of emergencies. NTPC - Simhadri emergency response plan to be prepared in line with this LMI taking care of the Legal requirements applicable this Station.

2.0 SUPERSEDED DOCUMENTS:

--- NIL---

3.0 REFERENCES :

COS-ISO-00-OGN/OPS/SAFT/002- Rev 0, July 2021- **Emergency Response Planning**

4.0 REVISION DETAILS:

---NIL---

5.0 INTRODUCTION:

NTPC has made a policy commitment to compliance with applicable legal and other safety and occupational health requirements that relates to its activities. Having in place an emergency response plan is a statutory requirement for most locations.

In addition to statutory requirements, in many others ways it is a prudent measure to have in place an ERP, for the protection of personnel and property of NTPC and the general public and for the recovery from emergencies and disasters in the shortest possible time.

Emergency Response Plan of NTPC - Simhadri shall provide both a management and an operational structure that identify the functional relationships that must exist to ensure effective identification, response, and recovery from an emergency.

6.0 ROLES & ACCOUNTABILITY :

HOPs shall be accountable to:

- Ensure that NTPC – Simhadri has in place an ERP before operational liability is transferred to the company.
- organize to validate adequacy of emergency action plans, at least twice a year.
- keep the Management apprised of any potential issues related to the location's emergency response, periodically.
- ensure a system is in place to account for the number of persons inside the project premises.
- organize to ensure all persons working within the premises are aware of emergency response requirements for their respective roles.
- arrange to identify emergency equipment under various functions/sections.

For areas where operations and maintenance functions are outsourced to third-parties, the designated Person accountable for the performance of such a project, where ever he/she is located, shall be accountable for the above role.

Head of Section (HOS) :

- Ensure periodic inspection of the emergency management equipment and facilities.
- Ensure persons in their respective Sections are aware of emergency response requirements for their respective Section.

Site Safety Coordinator:

- Organize for periodic exercise like table-top exercises and field mock drills, as advised by the HOP.
- Monitor availability of **persons** for all appointed emergency response roles and include the status in periodic reviews.
- Coordinate with local training manager for training on Emergency Response Plan of the location to cover all employees of NTPC – Simhadri.
- Review periodically the adequacy of the emergency management infrastructure as identified in the location ERP and maintain records.

All employees working at the location shall actively seek to understand emergency response requirements for their respective roles.

7.0 SYSTEM REQUIREMENT FOR EMERGENCY RESPONSE :

7.1 IDENTIFICATION OF EMERGENCY SCENARIOS :

Risk Assessments provide a starting point for identification of potential emergency scenarios. The location Emergency Response Plan (ERP) should be based on assessed hazards and risks and build-in responses that provide the flexibility to adapt to emergency scenarios regardless of their origin.

Each emergency is unique accordingly it becomes impossible to envision every potential event or combination of events that might contribute to, or result in, an emergency at the location. Annexure-1 provides some possible scenarios for different circumstances.

RESPONSIBILITY:

ALL HODs & HEAD OF O&M

7.2 EMERGENCY MANAGEMENT PLAN :

Emergency Response Plans should be in writing and cover all the procedures considered necessary to cater for all assessed emergency situations, be it natural or man-caused, and be available on site. In planning its emergency response, the site shall take account of the needs of relevant interested parties, e.g. government emergency services and neighbors. The following shall be considered while planning to tackle emergencies/disasters:

- Situations that can lead to emergencies or disasters are to be identified and listed. Emergency response plan should be developed for each type or category of emergency according to risk. All such assessed scenarios and plans are assembled together as the location's Emergency Response Plan. This is also legal requirement for most locations.
- The plan should be reviewed and endorsed by HOP at least once a year and records maintained.
- The plan shall identify the specific roles, responsibilities and action to be taken in the event of an emergency. All such roles & responsibilities shall be part of training in emergency response.
- Planned routes should be established for easy access to medical facilities/muster points. Awareness of planned routes shall be part of training in emergency response.
- Include a system to account for the number of persons inside the project premises so that no one is left behind in, or exposed to a potentially dangerous condition or unaccounted for.
- Provisions made should include assessments of facilities requirements that can arise out of emergencies. These assessments should include capacities available with the location's internal facilities and access arrangements for making available external services like the emergency referrals to the doctors/hospitals, numbers of beds available at different nearby hospitals, agreements with such facilities, aspects like possible logistics requirements to shift

persons in trauma to such nearby hospitals, and how such arrangements can be tested for effectiveness.

This assessment of facilities requirements to cater to emergency would become very critical for smaller locations where not much internal infrastructure may be available and/or remote locations, where efficient logistics can decide between life or death of persons.

- The location evacuation plan should make provision for the physically challenged people that work in the site.
- Staff involved in the event of emergency should be adequately informed & trained in the procedures. Records of all such training shall be maintained.
- The EMP should be introduced to all employees as part of the location induction program either when joining for the first time with the company or joining on a transfer from another location.
- The EMP shall also identify communication requirements to affected interested parties such as local community and emergency services, as appropriate.
- The site shall periodically test its ERP to respond to emergency situations; involving relevant interested parties as appropriate, at a frequency complying with legal requirements or, higher.
- The plan should be reviewed and revised if necessary, after periodic mock exercises or, any actual occurrence of emergency incident.

Emergency Response Plan shall be prepared inline with Annexure -2 and made available to all.

RESPONSIBILITY:

ALL HODs

7.3 EMERGENCY COMMUNICATION :

The Emergency Response Plan should include a communication matrix with single-point contacts and back-up roles for engaging with regional/local emergency and healthcare authorities.

The location should identify the relevant parties, which it may need to consult with, as part of emergency response like local administration (e.g., district magistrate), regulatory bodies (e.g. Dept. of Factories, Dist. Medical Officer), or emergency services (e.g. police, hospitals, fire services), those who may be affected by the location's emergencies (neighborhood populace).

Gaps in critical communication with external or interested parties can hurt the image of the company. To mitigate such risk, there should be a designated point of contact for all such safety & health related communications. This allows for appropriate information to be communicated in a consistent manner. This can be especially important in emergency situations where regular updates may be needed, and a wide range of questions may need to be answered.

The communication matrix should be included in the location's Emergency Response Plan and updated in the annual review or, if changes to significant roles has taken place due to transfers, etc.

RESPONSIBILITY:

HEAD OF SAFETY, CMO & HEAD OF HR

7.4 EMERGENCY RESPONSE TEAMS :

- HOP shall ensure Emergency Response teams such as search-and-rescue team, first-aid team, spill control team, etc., are set up at the location, to cope with the assessed emergencies and to facilitate implementation of relevant emergency response plans.
- Emergency services teams should be familiar with potential safety, health and environmental risks encountered in their roles and where applicable, be aware of the proper use of personal protective equipment and their limitations. Towards achieving that, emergency services teams should be trained in the location's Emergency Response Plan in their respective roles and responsibilities and be involved in regular emergency drills. Refresher training shall be imparted to the emergency response teams' members at least annually. Training records shall be maintained.
- Alternate roles should also be appointed as backup for Emergency team leadership roles.
- Performance of such back-up roles should be tested in active control for handling mockdrills, table-top exercises to enable them to get to grips with the expectations of the role. Such drills shall be recorded.

RESPONSIBILITY:

HEAD OF SAFETY in consultation with HOP

7.5 EMERGENCY COORDINATOR :

- HOP is generally the designated role at every location to coordinate the emergency action plan.
- An alternate coordinator should also be appointed by the HOP, as backup.
- Such back-up roles should be prepared for handling mock drills, table-top exercises to enable them to get to grips with the expectations of the role and records maintained.

RESPONSIBILITY:

HEAD OF SAFETY in consultation with HOP

7.6 CONTROL CENTER/ EMERGENCY EQUIPMENT :

- This arrangement shall be as specified by law or, NTPC standard, whichever is higher.
- A designated control center fitted with appropriate communications equipment and/or other emergency equipment, plan, etc. should be established for handling major

emergencies based on assessed risks.

- An alternative control center should be identified in the event of the main control center being rendered inoperative/inaccessible.
- Where applicable, emergency power supply should be planned and available for emergency lighting, communication and equipment.
- Appropriate spill control equipment should be made available and ready for use to handle expected environmental emergencies.
- Consideration should be given to the special types of equipment that may be needed for the expected emergencies and regular check should be carried out on the equipment by the respective HOS, to test its state of readiness for action.
- These dedicated control centers should be equipped appropriately and, maintained in appropriate working order. An indicative list of such equipment is listed below:
 - Communication systems like portable loud-speakers.
 - Fire tenders.
 - Ambulances.
 - Portable fire-fighting equipment.
 - PPE, based on assessed risks.
 - Emergency power supply.
 - Emergency alarm systems.
 - Portable lighting arrangements.
 - Spill containment equipment.
 - Life jackets/buoys.
 - Rescue vessels with qualified operators, where required.

The physical location of any emergency response related equipment would be based on assessed risks.

RESPONSIBILITY:

HEAD OF SAFETY, HEAD OF OPERATION & HEAD OF MAINTENANCE

7.7 MAINTENANCE OF EMERGENCY EQUIPMENT :

Periodic inspection of the emergency equipment should be ensured by the relevant Head of Section and records maintained. Site Safety Coordinator should include status of such activities in periodic reports.

RESPONSIBILITY:

HEAD OF SAFETY, SECTION HEADS & HODS

7.8 HEAD COUNT OF PERSONS WITHIN THE PREMISES :

There would be certain emergency scenarios that would necessitate evacuation of all persons working inside the location's premises. An essential requirement for any such evacuation is the accurate accounting for the persons inside the premises. Normally a headcount is taken at the designated assembly points, after the workplace evacuation. This

helps identify the names and last known locations of anyone not accounted for and to initiate precise search-and-rescue ops. Any doubts on this number can lead to delays in rescuing any one trapped inside debris or launching unnecessary and dangerous search-and-rescue operations. HOP shall arrange to have a system that ensures 100% accounting for the number of persons inside the location's premises.

RESPONSIBILITY:

HEAD OF HR

7.9 EXERCISES :

Periodic exercises are required to be enacted to validate the soundness and the continued appropriateness of the planned emergency response. Such exercises can be in the form of mock drills in the field e.g., grass fires or table-top exercises e.g., complete evacuation of the project due to a disaster like boiler explosion triggering multiple fires to test relevant role's clarity in emergency response planning on aspects like mobilization of external agencies, logistics.

The frequency of such exercises shall meet or exceed the location's applicable legal requirements.

RESPONSIBILITY:

HEAD OF SAFETY, CMO ,HEAD OF HR & HEAD OF O&M

8.0 RECORDS:

Any record that is maintained as an electronic copy shall be in a format that cannot be edited. The following records shall be maintained:

1. HOP:
 - Annual review of the location's Emergency Response Plan;
 - Appointment records of backup role for Emergency Coordinator, and for the roles of emergency response teams and back-up roles for the same;
 - Evidence of a system that ensures 100% counting of persons inside the location's premises at any point in time.
2. All HOSs shall maintain:
 - Records of periodic inspections of emergency management equipment under their control;
 - Records of review of emergency response plan with all their respective Section

employees at least annually – applicable to all HOS, irrespective of their role in the location's emergency response plan.

3. Site Safety Coordinator:

- training records of Emergency Response team members in location emergency response plan with respect to their roles.
- training records of all staff on the location's emergency response plan.
- evidence of periodic exercises, associated observations, learning and where appropriate, corrective actions.
- evidence of compliance to legal requirements of the location on emergency response.

All records can be maintained in a central repository.

For areas where operations and maintenance functions are outsourced to third-parties, the designated Person accountable for the performance of such an area, where ever he/she is located, shall be accountable for ensuring a system of reporting and records at such projects under his/her control and have such records accessible for audit purposes.

RESPONSIBILITY:
AS MENTIONED ABOVE

9.0 AUDIT :

EVALUATION OF COMPLIANCE :

Evaluation of the compliance shall be carried out at least once annually. The compliance evaluation assessment/audit can be included as part of other compliance audits or assessments like technical compliance audits. The audit shall be a combination of verification of evidence in the form of records, verification of implementation in the field and interviews.

Records: Audit shall include verification of all the records detailed under 8.0. The project should be able to demonstrate that it has identified potential emergency scenarios based on risks.

The copy of this LMI shall be available with all Departments for compliance

Responsibility
All HODs.

Implementation: Auditors shall validate the evidence of a system for verifying the headcount of all persons working at any given point in time within the location's premises. Auditors shall verify maintenance processes for the equipment identified as part of emergency response planning.

Field audit: shall be in the form of random assessments. Auditors shall check at random any two emergency response related systems like emergency siren, life jackets, fire tender . Check their state of readiness/availability and record the same in their audit report.

Interviews: auditors shall interview at least two HOS', at least five employees and at least five outsourced persons regarding their awareness of the location's emergency response plan and their role in that; awareness of emergency contact numbers; awareness of planned emergency routes; awareness of how to reach nearest Assembly Points - respective HOS shall be accountable for demonstrating compliance to such awareness requirements. Activities verified and persons interviewed shall be part of the audit report.

For projects where operations and maintenance functions are outsourced to third-parties, the designated Person accountable for the performance of such a project, where ever he/she is located, shall be accountable for ensuring periodic audits of the system at such projects under his/her control and maintain records.

RESPONSIBILITY:
All HODs

10.0 REVIEW:

The Head (EEMG) shall arrange for review of this LMI by the Head(safety) as and When required, or every two years.

ANNEXURE 1 - SAMPLE EMERGENCY SCENARIOS

ACTUAL SCENARIOS ASSESSED AS PER 7.1

	NTPC - Simhadri
	Explosion in boiler – need to handle 150 bodies in severe trauma.
	Fire in Coal handling plant.
	Simultaneous fire in all fuel oil tanks.
	Fire and Explosion in Hydrogen storage/handling area/ uncontrolled Hydrogen release
	Fire and explosion in TG .
	Fire in track hopper , people trapped.
	Fire in Cable Gallery
	Fire in Switch Yard with domino effect
	Earthquake – leading to collapse of structures and entrapment of people.
	Fire in Coal Conveyor with potential to cause structural collapse
	Leakage of Acids (Sulfuric acid or Hydrochloric Acids) from acid storage area.
	Cyclone expected with 300kmph wind speed – project was designed for max wind speeds of 250kmph.
	Solar Plant- O&M Contractor absconded due to local issues - project has become unmanned.
	Breach of Ash dyke/Raw water reservoir

ANNEXURE 2 - OUTLINE OF EMERGENCY RESPONSE PLAN

A broad outline is given below for the structure of the Emergency response Plan preparation. All the possible scenarios and action plan may be covered in the Disaster management Plan.

1. Purpose
2. Objective
3. Level of Emergency
4. Major Hazard Potential
5. First Information & declaration of emergency
6. Evacuation & Assembly points
7. Emergency Control centre
8. Emergency/Disaster management Organization
9. Training & Testing of plan